CASE NO. QX · QUANTUM INVESTIGATION UNIT

ECHO

An echo, not a voice.

Why Dr. Mahmud’s experiment never sent a message, and why its trigger was not random.

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THE VERDICT

Zara mistook correlation for communication, and someone made sure ECHO fired.

High confidence · ~90%

ECHO could not send a message faster than light

Moderate confidence · ~60%

The trigger was a chosen setting, not chance

Low confidence

Who set it: the physics alone cannot name them

THE EVIDENCE

Five modules, five facts

Module	What I measured	What it proves
01	H, H → always 0 · vault code 101	Interference: “random-looking” can be certain
02	Seed ry(1.159): P(1) = 0.30	The trigger coin was rigged, not fair
03	Bob stays 0.50 whatever Alice does	Correlation, not communication
05	Majority vote → “ECHO”	Message recovered, if ≤ 1 flip per triple
06	Teleport: 100% with bits, 51% without	No message without a classical channel

MODULES 01 + 05

Getting inside, and recovering her last word

01 · The Gatekeeper

`H, H → {'0': 2000}`

After one H the qubit looks random. The second H makes the paths to 1 cancel. Certain, not random.

`x(0), x(2) → {'101': 2000}`

05 · The Damaged Message

ECHO

Majority vote fixed 20 of 32 triples. Two flips in one triple would silently fool it.

MODULE 02 · THE HONEST COIN

The “random” seed that fired ECHO was rigged

30%

chance of 1 · a fair coin gives 50%

`seed.ry(1.159) → P(1) = 0.301`

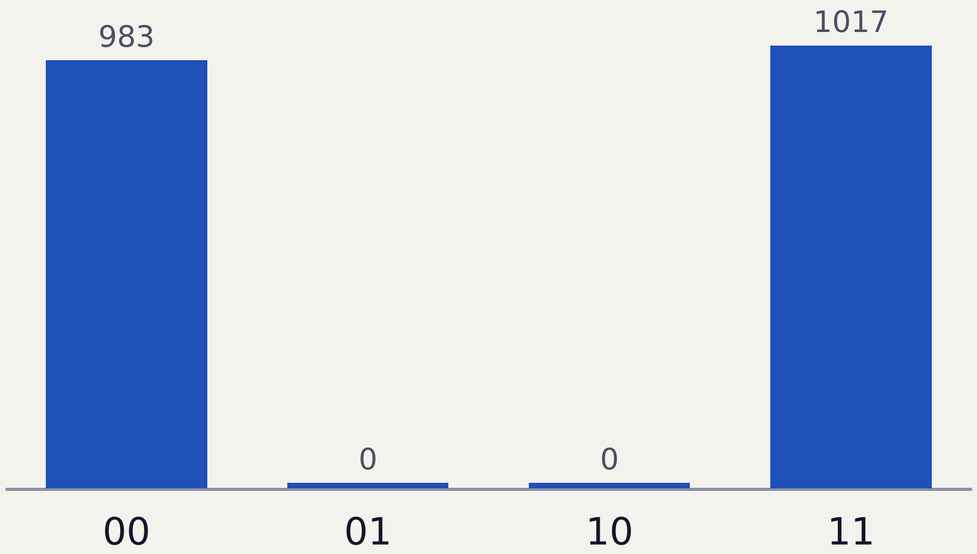
`fair coin needs ry(π/2 = 1.571)`

10,000 shots · z ≈ −40. A fair coin passing |z| = 3 is already unlikely. This is not bad luck.

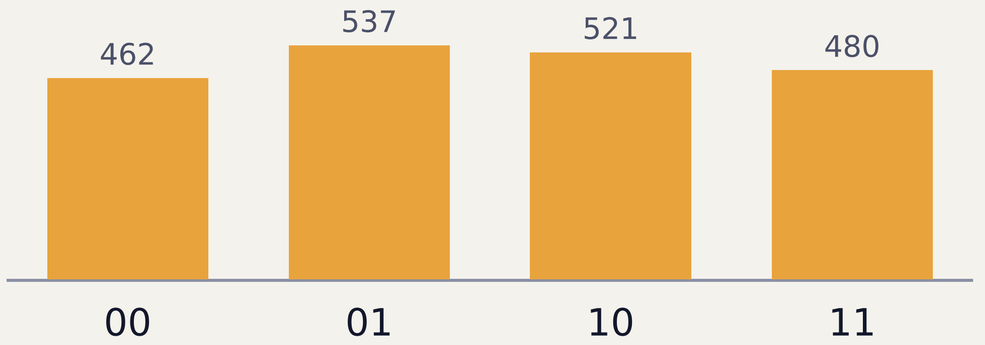
Only ~60 shots are needed to spot a bias this big.

Always agreeing proves nothing. Measuring the other way does.

Entangled pair · still agrees



Pre-agreed secret · goes random



Both give only 00 / 11 when measured normally. Shown: H applied to both qubits first (2,000 shots).

THE CRUX

Whatever Alice does, Bob still sees 50/50.

Alice does nothing	Alice applies H	Alice applies X	Alice measures
0.504	**0.504**	**0.504**	**0.508**

Bob’s chance of reading 1 never moves, so he cannot read any message. The Bell test gives S = 2.82 (above the classical limit of 2): the link is genuinely quantum, and it still carries nothing.

Teleportation works only when two ordinary bits arrive

100%

receiver correct, with the 2 bits

51%

without them: a coin toss

Message angle	Target P(1)	With bits	Without
0	0.00	0.00	0.51
π/3	0.25	0.26	0.50
2π/3	0.75	0.75	0.50
π	1.00	1.00	0.50

Without the bits Bob gets 0.50 for every message, so his qubit alone carries nothing. The bits travel no faster than light. **ECHO was never faster than light.**

THE COMPETING THEORY

Could it all have been an honest mistake?

The alternative

Nobody helped her. Zara sincerely misread correlation as a signal, and 1.159 was a calibration error.

Why I prefer mine

- 1.159 is far from 1.571: a 20-point bias
- A log was deleted (Sana)
- A warning report went missing (Bilal)
- ARIA: “no unauthorised command”, so someone authorised it

HONEST LIMITS

What my evidence cannot prove, and what I’d test next

Cannot prove

- Who set the seed: the physics shows deliberate, not by whom
- That each triple had at most one flip: the “ECHO” decode depends on it
- Real hardware: all runs were on a noise-free simulator

Next experiment

Check the seed’s configuration history.

Ever set to π/2, then changed to 1.159 → tampering, and the change log names the account.

Always 1.159 → calibration mistake more likely.

CASE CLOSED

Entanglement links the answers. It never carries the question.

github.com/huzaifaguru/echo-quantathon

Syed Mohammad Huzaifa · Tools: Qiskit 2.x + Aer simulator · AI assistance (Claude) acknowledged in the case report